

PEM Fuel Cell Stack — Experimental Results

TR-Current Experiments: $3.3\ \Omega$ and $6.8\ \Omega$ loads

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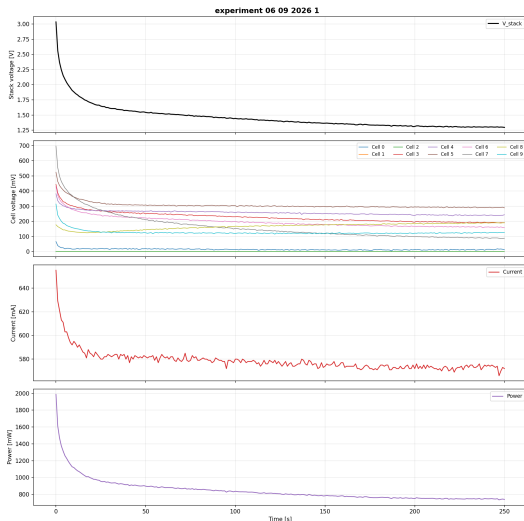
GIPSA-lab — Université Grenoble Alpes

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Outline

- 1 Experimental Data
- 2 Current Response Time Analysis
- 3 Voltage Response Time Analysis
- 4 Conclusions

Experiment 1 — $3.3\ \Omega$ Load



Conditions

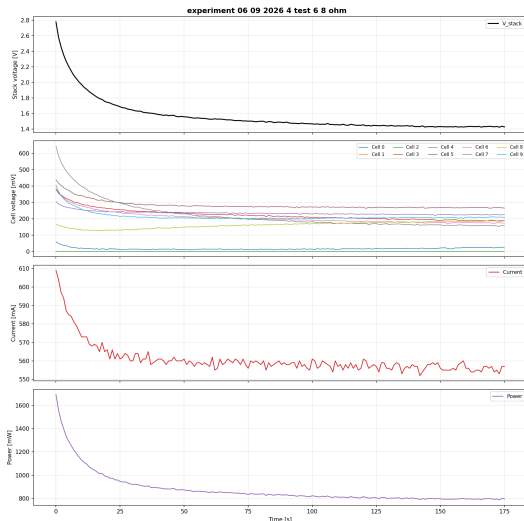
$R = 3.3\ \Omega$, recorded 9 June 2026

Channels (top to bottom)

- 1 Stack voltage V_{stack} [V]
- 2 Cell voltages $v_0 \dots v_9$ [mV]
- 3 Current I [mA]
- 4 Power $P = V_{\text{stack}} \cdot I$ [mW]

Cells 1 & 2 read $\approx 0\ \text{mV}$ — hardware issue.

Experiment 4 — $6.8\ \Omega$ Load



Conditions

$R = 6.8\ \Omega$, recorded 9 June 2026

Channels (top to bottom)

- ① Stack voltage V_{stack} [V]
- ② Cell voltages $v_0 \dots v_9$ [mV]
- ③ Current I [mA]
- ④ Power $P = V_{\text{stack}} \cdot I$ [mW]

Data starts at the current peak.
Same dead cells observed.

Current Response — Exponential Decay Model

After resistor connection the current follows a decaying step:

$$I(t) = I_{ss} + \Delta I \cdot e^{-t/\tau}$$

- I_{ss} : steady-state current (mean of last 20 % of samples)
- $\Delta I = I(0) - I_{ss}$: step amplitude
- τ : time constant (least-squares exponential fit)

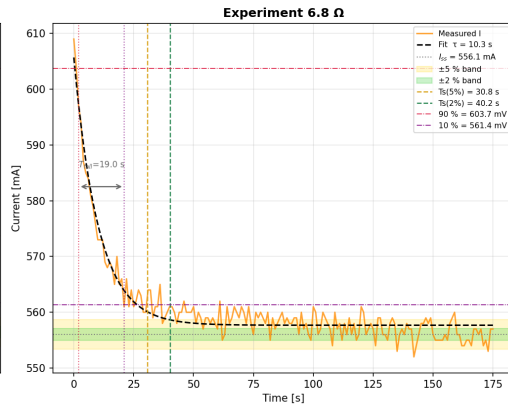
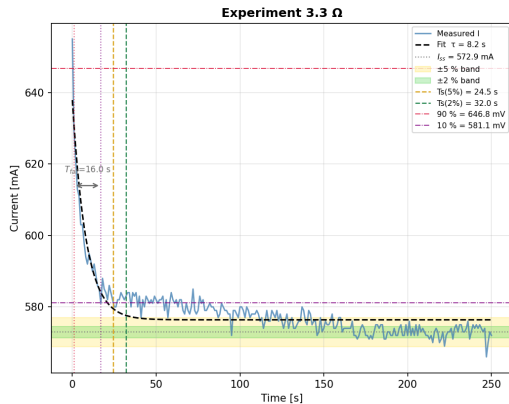
Settling times derived analytically from the fit:

$$T_{s,5\%} = -\tau \ln(0.05), \quad T_{s,2\%} = -\tau \ln(0.02)$$

Fall time 90 % $\rightarrow$ 10 % from the measured signal directly.

Response Time Results

Current Response Time Analysis



Response Time Metrics — Current Summary

Metric	3.3 Ω	6.8 Ω
Steady-state current I_{ss} [mA]	572.9	556.1
Step amplitude ΔI [mA]	82.1	52.9
Time constant τ [s]	8.2	10.3
Settling time $T_{s,5\%}$ [s]	24.5	30.8
Settling time $T_{s,2\%}$ [s]	32.0	40.2
Fall time 90 % $\rightarrow$ 10 % [s]	16.0	19.0

Observation

The 6.8 Ω load draws less current ($\approx 25\%$ smaller ΔI) and responds $\approx 25\%$ slower ($\tau = 10.3$ s vs. 8.2 s), consistent with lower electrochemical activity at reduced current.

Voltage Response — Exponential Decay Model

After resistor connection the stack voltage also follows a decaying step:

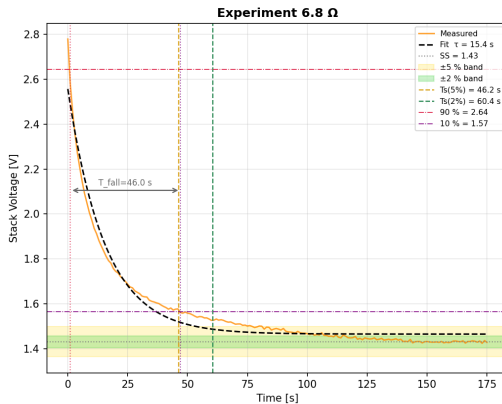
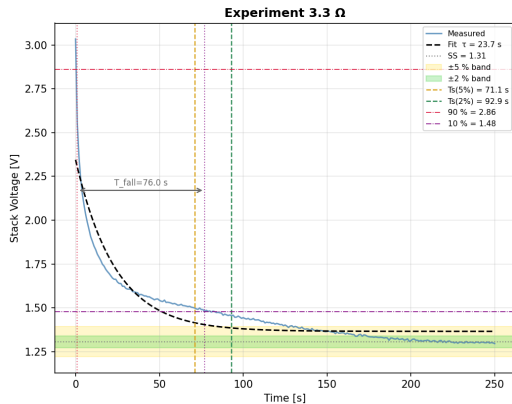
$$V(t) = V_{ss} + \Delta V \cdot e^{-t/\tau_V}$$

- V_{ss} : steady-state stack voltage
- $\Delta V = V(0) - V_{ss}$: voltage drop amplitude
- τ_V : voltage time constant (independent fit from current)

The voltage time constant is generally larger than the current time constant because the membrane hydration and gas pressures relax on a slower timescale.

Voltage Response Time Results

Voltage Response Time Analysis



Response Time Metrics — Voltage Summary

Metric	3.3 Ω	6.8 Ω
Steady-state voltage V_{ss} [V]	1.307	1.431
Step amplitude ΔV [V]	1.728	1.348
Time constant τ_V [s]	23.7	15.4
Settling time $T_{s,5\%}$ [s]	71.2	46.3
Settling time $T_{s,2\%}$ [s]	92.9	60.4
Fall time 90 % $\rightarrow$ 10 % [s]	76.0	46.0

Observation

The 3.3 Ω load produces a larger voltage drop ($\Delta V = 1.73$ V) and a slower voltage response ($\tau_V = 23.7$ s vs. 15.4 s), while the current settles faster. This indicates that membrane and gas-management dynamics dominate the voltage relaxation.

Conclusions

1 Data acquisition and preprocessing

Automatic resistor-connection detection and time trimming pipeline implemented in Python. Two clean datasets produced.

2 Current response

Exponential decay fits: $\tau_I = 8.2 \text{ s}$ (3.3Ω) and $\tau_I = 10.3 \text{ s}$ (6.8Ω). Current settling time $T_{s,2\%}$: 32 s and 40 s respectively.

3 Voltage response

Voltage relaxes more slowly: $\tau_V = 23.7 \text{ s}$ (3.3Ω) and $\tau_V = 15.4 \text{ s}$ (6.8Ω). Settling time $T_{s,2\%}$: 93 s and 60 s. The membrane and gas-management dynamics dominate voltage relaxation.

Next steps: Use the measured step responses to characterise stack dynamics and design a closed-loop controller.

Thank you

Questions?